Name: _____ Reg #: _____ Section: _____

National University of Computer and Emerging Sciences, Lahore Campus

Course:	**Natural Language Processing**	**Course Code:**	**CS 535**
Program:	**MS(Data Science)**	**Semester:**	**Spring 2023**
Duration:	**35 Minutes**	**Total Marks:**	**19**
Paper Date:	**23-Feb-2023**	**Weight**	
Section:	**DS**	**Page(s):**	**4**
Exam:	**Quiz 1**		

Q1) You are given the following training corpus: [1 + 2 + 2 + 3 + 2 = 10 Marks]

```
<s> I am Sam </s>
<s> Sam I am </s>
<s> I am Sam </s>
<s> I do not like green eggs and Sam </s>
```

a) Calculate the probability of following test sentence. Include </s> in your counts just like any other token.

<s> I am like Sam </s>

i. Unigram Model

ii. Bigram Model

iii. Trigram Model

iv. Bigram language model with linear interpolation. λ_1 = bigram weight, λ_2 = unigram weight, λ_1 = 0.7, λ_2 = 0.3

b) Calculate perplexity of test sentence using bigram model

Q2) A dictionary can be used to identify spelling errors. How does language modeling help in task of spelling correction? [3 Marks]